

**KNEE OSTEOARTHRITIS PREDICTION AND
PROGRESSION USING MULTI-MODAL DEEP
LEARNING.**

Project ID: 25-26J-112

Jayasinghe J.M.N.S

IT22582942

BSc (Hons) in Information Technology Specializing in Data Science

Department of Information Technology
Faculty of Computing

Sri Lanka Institute of Information Technology
Sri Lanka

April 2026

**KNEE OSTEOARTHRITIS PREDICTION AND
PROGRESSION USING MULTI-MODAL DEEP
LEARNING.**

Project ID: 25-26J-112

Jayasinghe J.M.N.S – IT22582942

Supervisor - Ms. Jenny Krishara

BSc (Hons) in Information Technology Specializing in Data Science

Department of Information Technology

Faculty of Computing

Sri Lanka Institute of Information Technology

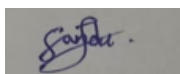
Sri Lanka

April 2026

DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Also, I hereby grant to Sri Lanka Institute of Information Technology, the non exclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

Name	Student ID	Signature
Jayasinghe J.M.N.S.	IT22582942	

The supervisor/s should certify the dissertation with the following declaration.

The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

Signature of the Supervisor

Date

.....
(Ms. Jenny Krishara)

.....

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my dissertation supervisor, Ms. Jenny Krishara, and co-supervisors, Ms. Thamali Dassanayake and Ms. Wishalya Vanshanee Tissera of the Faculty of Computing, for their invaluable guidance and continuous support throughout this research journey. Their insightful feedback and constructive suggestions have greatly contributed to enhancing the quality and depth of my work.

I would also like to extend my sincere appreciation to Dr. Harsha Perera, External Supervisor and Doctor at Colombo North Teaching Hospital, Ragama, for his valuable insights and clinical guidance throughout this study. His expertise was particularly helpful in shaping the clinical and diagnostic aspects of this research.

My heartfelt thanks go to the academic and administrative staff of the Faculty of Computing for their consistent encouragement and support during my academic career. Their dedication to teaching and research has been a source of inspiration.

I am deeply appreciative of the support and collaboration of my teammates. The discussions, knowledge sharing, and teamwork significantly enriched my understanding of the subject matter and contributed positively to the outcome of this research.

Finally, I would like to thank all the individuals who participated in this study and shared their valuable experiences and perspectives. Their input has been instrumental in shaping the direction and findings of this research.

ABSTRACT

KOA is a degenerative joint condition and a major source of disability in the world, especially within the elderly and vulnerable populations. Timely diagnosis of KOA is important for early intervention. However, traditional diagnostic methods, including X-rays and MRI, mostly detect the disease in its later stages and can be expensive and inaccessible in low-resource environments. This study proposes a predictive machine learning model that uses non-imaging data such as age, sex, clinical symptoms and laboratory markers to predict the occurrence of KOA in its early stages.

This research uses a dataset of real patient data obtained from hospital visits, augmented with synthetic data for improved model training. The data was preprocessed to handle missing values, normalize, encode and create new features. The class imbalance issue was tackled using the SMOTE. Several supervised machine learning algorithms, such as Logistic Regression, Random Forest, Gradient Boosting, KNN, and XGBoost, were trained and validated using accuracy, precision, recall and F1-score metrics.

The Gradient Boosting model showed superior performance, with good accuracy and generalization. Furthermore, an XAI system using SHAP was incorporated to explain the importance of features and the reasoning for individual patients, improving the transparency of the model and its interpretation in a clinical context.

The developed system provides a non-invasive, scalable and cost-efficient approach for early KOA prediction, especially when imaging resources are scarce. This study explores the use of structured clinical data and explainable machine learning methods to build clinical decision-support systems that can enhance early detection, patient care, and health equity.

Keywords: Knee Osteoarthritis, Demographic data, Clinical data, Biomarker data, Machine Learning

TABLE OF CONTENTS

DECLARATION.....	iii
LIST OF FIGURES	vi
LIST OF TABLES	viii
LIST OF ABBREVIATIONS	ix
1. INTRODUCTION.....	1
1.1 Background	1
1.2 Literature Review	3
1.3 Research Gap	5
1.4 Research Problem	6
1.5 Research Objectives.....	8
1.5.1. Main Objective	8
1.5.2. Sub-Objectives.....	8
2. METHODOLOGY	9
2.1 Methodology	9
2.1.1. System Overview	9
2.1.2. Data Collection and Preprocessing.....	10
2.1.3. Functional, Non-Functional Requirements	14
2.2 Commercialization Aspect of the Product	16
2.3 Testing and Implementation	17
2.3.1. Implementation	17
2.3.2. Testing.....	21
3. RESULTS	24
3.1. Results	24
3.2. Research Findings.....	26
3.3. Discussion.....	28
4. SUMMARY OF INDIVIDUAL CONTRIBUTION – JAYASINGHE J M N S (IT22582942).....	31
5. CONCLUSION	32
REFERENCES.....	34
GLOSSARY.....	36

LIST OF FIGURES

Figure 1– System overview diagram.....	9
Figure 2– Collected dataset to train the model.....	11
Figure 3 – Filling missing values in biomarkers	12
Figure 4 – Standardizing numerical features	13
Figure 5 – Implementation of XAI using SHAP.....	18
Figure 6 – Model Implementation in the backend	19
Figure 7 – Frontend Dashboard	20
Figure 8 – Learning curve of the final model	25
Figure 9 – Confusion matrix of the final model.....	26

LIST OF TABLES

Table 1: Comparison of former research. 6

LIST OF ABBREVIATIONS

Abbreviation	Description
KOA	Knee Osteoarthritis
AI	Artificial Intelligence
BMI	Body Mass Index
COMP	Cartilage Oligomeric Matrix Protein
CRP	C- Reactive Protein
CT	Computed Tomography
CTX	Cerebrotendinous Xanthomatosis
EHR	Electronic Health Record
IoT	Internet of Things
ML	Machine Learning
MRI	Magnetic Resonance Imaging
SaaS	Software as a Service
SHAP	Shapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
VAG	Vibroarthrography

1. INTRODUCTION

1.1 Background

The rapid progress in artificial intelligence and machine learning over the past few years has revolutionised the field of healthcare, especially when it comes to prediction and early detection of diseases. The growing access to structured data about patients, such as demographic information, clinical history, and laboratory measurements of disease biomarkers, has opened up possibilities for the development of smart systems to help clinicians in early detection of diseases. Such data-driven solutions are particularly crucial in diseases where early diagnosis can lead to better patient outcomes and lower the risk of complications. One such disease is KOA, which is a significant public health burden because of its prevalence and impact on patients' quality of life.

KOA is a progressive joint condition that involves deterioration of cartilage, causing pain, stiffness, swelling and limited movement. It is a major cause of disability globally, affecting the elderly, and people with risk factors including obesity, history of joint trauma, and a lack of physical activity. The ageing population in many countries means that the impact of KOA is likely to rise, placing greater demands on health services. Due to the progressive nature of the disease, early diagnosis is essential to slow down the disease, alleviate symptoms and improve quality of life.

KOA diagnosis has historically been based on imaging technologies like X-rays and MRI. These techniques are good at detecting structural abnormalities in the knee such as cartilage degeneration, joint space narrowing and bone deformities. But a key drawback of such imaging techniques is that they often diagnose the disease at a late stage, when irreversible damage has already been done. Moreover, radiology procedures can be expensive, require highly specialized equipment, and require interpretation by experts, which can limit their availability in resource poor settings and rural areas. This leads to a delay in diagnosis and limits the benefits of early interventions.

Because of these challenges, there is a growing demand for other methods that can aid in the early diagnosis using readily accessible and non-invasive information. Age, sex, BMI, medical history and symptoms have long been acknowledged as risk factors for

KOA. In addition, laboratory-based biomarkers such as inflammatory markers and other blood-based markers offer insights into the physiological processes that underpin the disease. While all of these have important roles, they are usually considered separately in clinical practice, reducing their capacity to yield a holistic view of disease risk.

This limitation can be overcome by using ML to merge and model multiple, disparate data sources in a single predictive model. ML approaches can automatically learn non-linear relationships between features and disease outcomes, which traditional statistical methods cannot. This is especially valuable in the medical field, where the development of a disease is influenced by multiple inter-related variables. These methods can be used to build models capable of detecting early signs of KOA, even before they're detectable by X-rays.

Our research aims to build a machine learning prediction system that uses demographic, clinical and biomarker information to predict KOA. The system uses a range of supervised learning techniques such as Logistic Regression, Random Forest, Gradient Boosting, KNN and XGBoost to compare different models and determine the best approach. The models are trained, validated and fine-tuned to enhance their predictive accuracy and generalization ability. The chosen model, Gradient Boosting, exhibits high accuracy and robustness on different patient data.

A major component of this work is the use of XAI methods to overcome one of the critical issues with machine learning in healthcare: interpretability. In healthcare, it's not enough for the system to make accurate predictions; it's crucial that it also explains its reasoning. Through techniques like SHAP, the system offers insights into the influence of each feature on the predictions, enhancing transparency and understanding. This builds confidence in the clinicians and facilitates informed decision-making, improving the readiness of the system for clinical use.

The proposed approach also offers a cost-effective and scalable solution. The system uses readily available patient data, obviating the need for extra diagnostic procedures and costly tests. This is especially beneficial in resource-poor settings, where access to sophisticated imaging equipment can be lacking. Additionally, the system can be integrated with digital health platforms, allowing for ongoing risk assessment of patients.

Overall, this research offers a data-driven approach to KOA prediction, overcoming the drawbacks of conventional diagnostic techniques. Through the integration of machine learning, clinical data and explainable AI, the system offers a valuable solution for early diagnosis and clinical decision-making. Predicting KOA using non-imaging data not only increases the accessibility of this technology but also encourages preventative care, leading to improved patient outcomes and more efficient health care systems.

1.2 Literature Review

The prediction of Knee Osteoarthritis based on demographic, clinical and biomarker data has emerged as a significant area of research as the earlier the disease is detected and the sooner it is treated, the less damage is caused to the joints in the long-term. In contrast to imaging alone methods, structured patient data has the potential to measure risk factors, symptom burden and biological processes involved in the development of KOA. Recent research has thus investigated machine learning approaches to predict the relationship between patient attributes and KOA status, particularly, risk factors, inflammatory biomarkers, and externally validated prediction models. Nonetheless, there are still gaps in the research of incorporating a variety of clinical variables into non-imaging decision-support tools that can be practically deployed.

Recently, Chen et al. [1] designed and tested machine learning-based predictive models of KOA diagnosis with biomarkers on a large data set of 2,594 samples of clinical data. Their study showed that biomarker-based models could be effective in differentiating KOA cases and that structured laboratory data have the potential to be diagnostic. Likewise, Kokkotis et al. [2] explored the issue of risk factors and machine learning-based prediction models on KOA patients demonstrating that demographic and clinical indicators can be employed to construct effective classifiers. The current studies are a strong encouragement of using structured tabular data to predict KOA, albeit focusing more on the development of models and predictive performance as opposed to a larger-scale incorporation of regular clinical screening variables within a hospital-based decision-support environment.

Lourido et al. [3] furthered this path by suggesting a clinical model which factored protein biomarkers to forecast radiographic KOA through Osteoarthritis Initiative data. Their future design demonstrated that the integration of both clinical factors and

biomarker data enhances predictive power, and the significance of multimodal structured data in KOA assessment. Similarly but more specifically, Long et al. [4] studied KOA risk prediction among the injured population, with previous lower-limb injury being found to be a valuable predictor of subsequent osteoarthritic alterations. The results are very applicable in the current component since they affirm that clinical history and symptom-related variables can significantly be used to model KOA risk, in addition to mere demographic profiling.

McCabe et al. [5] also added to this field by creating and externally validating diagnostic and prognostic models of KOA with baseline clinical data of large cohort studies. Their research is particularly valuable as it has focused on external validation, indicating that the performance of models needs to be checked outside of the initial training population. Simultaneously, Lee et al. [6] also examined the extended application of artificial intelligence in KOA diagnosis and reported that in spite of the potential of AI models, numerous studies continue to experience low levels of external validation, bias in the data, and poor clinical translation. This shows that predictive performance alone needs to be very strong, models do not just have to be predictive, but also have to make sense, be clinically relevant, and they need to be based on variables that can be realistically gathered in actual practice.

The other contribution is a cross-sectional study by Schadler et al. [7], on blood biomarkers and body mass index conducted on candidate diagnostic biomarkers in relation to obesity and end-stage KOA. Their results emphasize the need to combine metabolic and inflammatory biomarkers with anthropometric measurements like BMI. Collectively, the literature presented above evidences the definite advancement in KOA prediction based on structured data, specifically, machine learning and biomarker analysis. Nevertheless, most studies emphasize single sets of biomarkers, subpopulations, or well-controlled cohort data. The number of studies that deal with a practical development of a clinically applicable prediction framework integrating demographic, symptom-based, and biomarker characteristics that are gathered at routine hospital visits is low. The purpose of this research component is to fill this gap by creating a KOA presence prediction model based on real patient records and clinical variables easily available, therefore, facilitating the early and scalable decision support in the typical healthcare setting.

1.3 Research Gap

KOA is highly influenced by clinical conditions, demographic variables and biomarkers. The current studies address mainly either the demographic data or clinical variables, however, small number of researches have been done regarding machine learning methods in various conditions. The study will combine clinical, demographic and biomarker data to come up with improved prediction on KOA.

To achieve diagnostic accuracy and early risk identification, several studies have examined the prediction of KOA by using different clinical, demographic, and biomarker information and applying different machine learning algorithms.

W. Chen et al., 2025 – Identification of biomarkers for knee osteoarthritis through clinical data and machine learning models

In order to forecast the risk of osteoarthritis in the knee, this study examined clinical and biomarker data. Their method revealed important predictors that influence KOA and showed that the random forest model outperformed other algorithms in terms of diagnostic accuracy.

G. Schadler et al., 2022 - The association of blood biomarkers and body mass index in knee osteoarthritis: a cross-sectional study

This study examined the association between the serum biomarkers and BMI in patients with KOA. Using a cross-sectional design, the researchers determined the biomarker levels, including the adipokines and inflammatory markers, as well as BMI to study their combined influence on KOA progression. Results demonstrated a significant association between elevated BMI and increased levels of certain biomarkers, which has identified obesity not only as a mechanical risk factor, but also as a metabolic pathway to KOA.

L. Lourido et al., 2021 – A clinical model including protein biomarkers predicts radiographic knee osteoarthritis: a prospective study using data from the Osteoarthritis Initiative

This research developed a clinical-biomarker model to predict the future development of radiographic KOA. By analyzing protein levels together with biomarker and clinical variables, the researchers were able identify whether a person is having KOA or not.

L. S. Lee, P. K. Chan et al., 2022 - Artificial intelligence in diagnosis of knee osteoarthritis and prediction of arthroplasty outcomes: a review

In this paper, the application of AI approaches, specifically machine learning and deep learning, to diagnosing KOA and in predicting outcomes following knee arthroplasty was reviewed. The authors reviewed how imaging (x-rays, MRI) and clinical features have been applied in AI models to characterize KOA severity, determine risk factors, and predict progression. They also highlighted the role of AI in surgical predictions in the form of recovery level, as well as the prediction of revision surgery after knee replacement. Reported models showed encouraging levels of accuracy both for diagnosis and for prognosis.

Feature / Aspect	W.Chen 2025	L.S.Lee 2022	L. Lourido 2021	Proposed System
Focus on demographic data	Yes ✓	No ✗	No ✗	Yes ✓
Focus on clinical data	No ✗	Yes ✓	Yes ✓	Yes ✓
Focus on Biomarkers	Yes ✓	No ✗	Yes ✓	Yes ✓
Use ML for predictions	No ✗	Yes ✓	No ✗	Yes ✓
Prediction based on a multi model	Yes ✓	No ✗	Yes ✓	Yes ✓

Table 1: Comparison of former research.

1.4 Research Problem

The main issue with the Knee Osteoarthritis care is the fact that it is quite difficult to detect the disease in its early stages and be certain of its existence. KOA is a progressive disease, which gradually increases with time and is often asymptomatic at its early stages. Existing methods of diagnosis like X-rays and MRI scans are highly applicable in confirmation of KOA. Nevertheless, these methods normally only notice the disease when it has already caused some noticeable structural damage to the joint. Consequently, diagnosis is usually made at later stages and this constrains the impact of early interventions which have the potential of slowing the progression of the

disease and hence enhance patient outcomes. Moreover, these imaging techniques are expensive and demand specialized equipment and interpretation by an expert and are not as applicable in screening or continuous monitoring.

The other major problem is that the majority of the current diagnostic techniques are mainly based on imaging data but other useful sources of patient information are not fully exploited. Demographic variables like age, sex, and BMI and clinical variables like the intensity of pain, stiffness of the joint, and prior injury history have been found to have significant roles in the occurrence and development of KOA. Moreover, biochemical and inflammatory biomarkers that are gained through laboratory tests are valuable clues to the mechanisms of underlying disease. These data sources are not often combined into a single predictive model although each of them separately helps to comprehend KOA risk. Such a non-integration leads to incomplete evaluations and a lower quality of diagnosis in general.

The issue is also exacerbated by the lack of resources and rural healthcare where sophisticated imaging technologies are frequently unavailable or limited. Such environments may also lead to delays in diagnosis as medical facilities are limited, and the disease may go untreated. The lack of available, data-driven predictive systems contributes to the increased risk of overlooking the indicators of the early onset of the problem, which will eventually lead to the decrease in mobility and the growth of pain, as well as an increased risk of long-term disability.

Moreover, the existing clinical decision-making models tend to be overly dependent on subjective evaluations and experience of the physician, which can result in inconsistencies and variability in diagnosis. In the absence of intelligent systems able to process vast amounts of patient data, it is difficult to detect underlying patterns and relationships that might be the signs of early-stage KOA.

To overcome these shortcomings, it is evident that an all-encompassing and data-driven solution should be developed that can combine demographic, clinical, and biomarker data into a single predictive model. With the help of machine learning methods, such a system can take into account the complex interrelationships between many factors and make valid predictions of KOA presence. This method can help clinicians make more informed decisions by allowing them to identify issues early, enhance the accuracy of diagnosis, and minimize the reliance of costly imaging

techniques. Finally, it provides a low-cost, scalable and available solution that can be implemented in both urban and rural healthcare environments to help in enhancing patient care and managing Knee Osteoarthritis.

1.5 Research Objectives

1.5.1. Main Objective

The main objective of this research is to implement a software base system to early prediction and identify the progression of KOA, by integrating clinical, demographic and biomarker data. Therefore,

1. Predicting the presence of KOA accurately through clinical, biomarker and demographic data using a multi-model.
2. Predicting the presence of KOA using X-ray and MRI images.
3. Identifying the severity level using clinical, biomarker and X-ray/MRI images.
4. Developing a IoT based knee health monitoring system using VAG signals

1.5.2. Sub-Objectives

To accomplish the overall goal of developing a data-driven KOA prediction system using clinical, demographic, and biomarker data, the research will address the following sub-objectives:

1. Multi-Model Data Integration:

Develop a unified system that collects and integrates demographic, clinical, and biomarker data to support accurate prediction of knee osteoarthritis.

2.KOA Prediction

Implement machine learning models capable of predicting whether an individual currently has or hasn't KOA

3. Incorporate Explainable AI (XAI):

Use XAI methods like SHAP to explain the importance of features and patient-specific insights, allowing clinicians to understand how the model reached a conclusion and ensuring the model is trusted.

2. METHODOLOGY

2.1 Methodology

2.1.1. System Overview

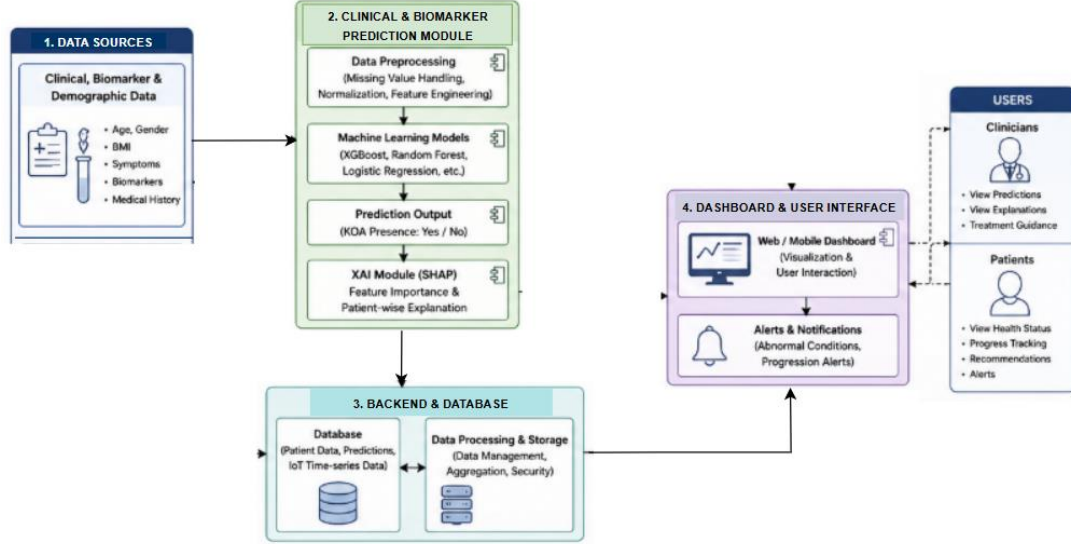


Figure 1– System overview diagram

This research proposes an innovative, data-centric system to assist in the early detection and tracking of KOA through the use of clinical, demographic and biomarker data. The system incorporates machine learning, XAI, and user-centered design to deliver precise, understandable, and user-friendly predictions. This provides an overview of the system's entire process, including data acquisition from multiple sources, prediction, explanation, storage and interaction, and the technical aspects and user experience of the system in the context of healthcare.

Data collection from multiple sources is the first step in the system, capturing a holistic view of patient health. These include structured clinical, demographic, and biomarker data, including age, gender, BMI, symptoms, medical history, and lab test results.

After data acquisition, the system moves on to the clinical and biomarker prediction module, the heart of this part. Here, data preprocessing steps are performed, such as handling missing data, normalization and feature engineering. This ensures data quality, consistency, and compatibility with machine learning models. New features like BMI are created to improve model performance, and encoding methods are applied to convert categorical variables into numerical format.

The transformed data is then fed into a range of machine learning algorithms, such as Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. The models learn to predict the occurrence of KOA based on the input features. Following a rigorous search and optimization process, Gradient Boosting was chosen as the best model based on its higher generalization ability and capacity to model non-linear relationships in medical data. The model provides a binary prediction of KOA presence.

Unlike conventional models, this system incorporates an XAI module, based on SHAP. This component adds an element of interpretability by offering feature importance values and individualized explanations. Rather than provide an opaque outcome, the system shows how age, BMI, or a particular biomarker, for example, affects the decision-making process. This transparency is essential in healthcare, where it ensures clinicians' trust and aids decision-making.

The system includes a comprehensive backend and database management layer to handle data storage, processing and communication. It stores all user inputs, predictions, and past data, allowing for trend analysis and monitoring over time. The backend also aggregates data, preprocesses the data, and maintains data integrity and security. It has a modular design for easy upgrades and extensions.

Lastly, the system provides a dashboard and user interface, which can be accessed through a web or mobile interface. Here, clinicians can enter patient data, see the prediction results and interpret model explanations through visual representations. Furthermore, the system supports proactive intervention with alerts and notifications, such as warnings for abnormal conditions or disease progression.

In conclusion, the system proposed in this paper combines data collection from multiple sources, state-of-the-art machine learning algorithms, explainable AI, and a user-friendly interface to predict KOA. The emphasis on non-imaging data and explainability makes the system a feasible, scalable, and affordable approach to early diagnosis and clinical decision making, especially in low-resource settings.

2.1.2. Data Collection and Preprocessing

In order to come up with a dependable and precise machine learning model to predict the occurrence of KOA, the first and most important step was to gather, structure and

pre-process a rich clinical dataset. The training and validation of the predictive models are based on this dataset such that the system is able to generalize to real-life clinical situations. As the proposed component is dedicated to non-imaging data, special attention was paid to gathering high-quality demographic, clinical, and biomarker data that will represent real-life conditions of patients.

The main data of this research was obtained by the real patient records that were obtained by multiple clinical visits in the National Hospital of Colombo. With the assistance of medical professionals, about 600 records of patients were collected and they were clinically authentic and relevant. These data contained important demographic variables like age, gender, and BMI and clinical variables like joint pain, stiffness, movement restrictions and knee injury history. Further on laboratory-based biomarker measurements such as inflammatory markers and blood test results were added, to give more detailed insights into the progression of the disease.

To increase the dataset size and diversity, more 400 synthetic data samples were created. These simulated records were developed on the basis of statistical distributions and trends of the actual clinical data, so that there is consistency with the actual medical conditions. The use of synthetic data served to enhance the robustness of the models, decrease overfitting, and resolve the possible class imbalance problems, especially when some types of patients were underrepresented.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Column 1	Patient Identific	age	gender	height	weight	occupation	physical_activity	living_environm	knee_pain	knee_pain_in_pi	stiffness_after_re	knee_injuries	swelling	difficulty_r
2	C1-6715	PID0471	48	male	171.6	62.3	yes	low	urban	no	4	never	no	no	standing lo
3	C1-5239	PID0206	42	male	165.9	65.4	yes	high	urban	yes	1	occasionally	no	yes	walking lor
4	C1-6290	PID0696	48	female	150.2	59	yes	high	rural	no	4	always	no	no	squatting
5	C1-3597	PID0534	29	female	152.9	82.8	yes	low	urban	no	3	frequently	no	no	standing lo
6	C1-9788	PID1002	50	female	155.4	69.1	no	low	rural	yes	4	always	yes	yes	squatting
7	C1-6524	PID0095	63	male	172.3	85	yes	low	urban	yes	5	always	no	yes	walking lor
8	C1-8962	PID0048	44	male	166.3	110.6	yes	high	urban	yes	4	always	no	yes	walking lor
9	C1-2335	PID0459	44	female	149.9	48.5	yes	moderate	rural	no	3	never	no	no	standing lo
10	C1-1165	PID0027	72	female	152.2	61.2	no	moderate	urban	yes	4	frequently	no	yes	squatting
11	C1-1700	PID0252	69	female	165.9	75.9	yes	moderate	rural	yes	5	occasionally	yes	no	climbing st
12	C1-3277	PID0052	67	male	176.5	82.3	yes	low	rural	yes	5	always	yes	yes	walking lor
13	C1-5132	PID0361	38	female	162.3	60.7	no	low	rural	yes	3	occasionally	no	yes	walking lor
14	C1-1679	PID0279	54	male	175.7	97.4	yes	low	rural	yes	5	frequently	yes	yes	squatting
15	C1-5598	PID0355	55	female	147.9	51.6	no	high	urban	yes	4	frequently	no	yes	squatting
16	C1-7798	PID1016	55	male	169	85.4	yes	moderate	rural	yes	1	occasionally	no	yes	climbing st
17	C1-4804	PID0388	76	female	166	94.9	yes	moderate	rural	yes	5	never	yes	no	walking lor
18	C1-8352	PID0365	41	male	170.8	108	yes	high	urban	no	5	frequently	no	no	walking lor
19	C1-3406	PID0833	53	female	156.7	75.2	no	moderate	rural	yes	2	always	no	yes	standing lo
20	C1-5684	PID0114	48	male	170.6	90.2	yes	moderate	urban	yes	4	frequently	no	yes	climbing lo
21	C1-1494	PID0183	56	male	164	63.6	yes	low	rural	yes	4	occasionally	no	no	climbing st
22	C1-9118	PID0735	37	female	157.2	66.1	yes	low	urban	no	2	never	no	no	squatting
23	C1-2738	PID0588	62	female	160.9	53.5	yes	high	rural	yes	1	occasionally	yes	no	squatting
24	C1-7656	PID0465	46	male	173.2	89.5	yes	high	rural	no	1	never	no	no	climbing st
25	C1-5557	PID0914	59	female	159.8	85.4	no	low	urban	no	3	never	yes	no	squatting

Figure 2– Collected dataset to train the model

After the data collection stage was fulfilled, extensive preprocessing pipeline was introduced in order to guarantee the quality of data, its consistency, and its

appropriateness to machine learning algorithms. As a preliminary step, non-informative attributes were eliminated, including patient identifiers and free-text fields, which do not add value to predictive performance and can be a source of noise. Numerical features with missing values were treated with median imputation, since this method reduces the influence of outliers, and does not alter the overall data distribution.

```
#handling missing values in numerical columns

def clean_numeric(col):
    df[col] = df[col].astype(str)
    df[col] = df[col].str.replace("mm", "", regex=False)
    df[col] = df[col].str.replace("MM", "", regex=False)
    df[col] = df[col].str.replace(" ", "", regex=False)
    df[col] = df[col].str.replace("\u200b", "", regex=False) # zero-width space
    df[col] = pd.to_numeric(df[col], errors='coerce')

# Clean and fill FBC
clean_numeric('fbc')
df['fbc'] = df['fbc'].fillna(df['fbc'].median())

# CRP
clean_numeric('crp')
df['crp'] = df['crp'].fillna(df['crp'].median())

# CS (Calcium Serum)
clean_numeric('cs')
df['cs'] = df['cs'].fillna(df['cs'].median())

# ESR
clean_numeric('esr')
df['esr'] = df['esr'].fillna(df['esr'].median())

# WBC
clean_numeric('wbc')
df['wbc'] = df['wbc'].fillna(df['wbc'].median())

# RF
clean_numeric('rf')
df['rf'] = df['rf'].fillna(df['rf'].median())

# Cholesterol
clean_numeric('cholesterol')
df['cholesterol'] = df['cholesterol'].fillna(df['cholesterol'].median())

# FBS
clean_numeric('fbs')
df['fbs'] = df['fbs'].fillna(df['fbs'].median())
```

Figure 3 – Filling missing values in biomarkers

Categorical variables were transformed into numerical representations to allow compatibility with the models. Label encoding was used to encode binary features, and one-hot encoding was used to encode multi-class categorical variables into multiple

binary columns. This transformation made sure that all features would be well used by machine learning models without creating unintentional ordinal relations.

Biomarker values are usually different in different records in terms of units and measurement scales. Thus, the standardization of units and the absence of inconsistencies were carried out in a data cleaning step. All the numerical characteristics such as the values of the biomarkers were then standardized using z-score. This guarantees that features are centered about an average of zero with a standard deviation of one so that features that are larger do not prevail in the learning process.

```
#standardize numeric columns

numeric_cols = [
    'age', 'height', 'weight',
    'knee_pain_in_past_week',
    'fbs', 'wbc', 'platelets', 'cs',
    'cholesterol', 'crp', 'esr', 'rf', 'fbc', 'BMI'
]

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train[numeric_cols] = scaler.fit_transform(X_train[numeric_cols])
X_test[numeric_cols] = scaler.transform(X_test[numeric_cols])
```

Figure 4 – Standardizing numerical features

Additional meaningful attributes were also derived using feature engineering. Indicatively, height and weight data were used to calculate BMI which gives a valuable risk indicator of KOA. These learnt characteristics helped to enhance the predictive performance of the model.

After preprocessing, the data was split into training and testing samples with the help of the stratified sampling method to preserve the ratio of KOA-positive and negative cases. SMOTE was also used to further treat the training data to correct class imbalance. This method is used to produce artificial samples of the minority class to make sure that the model is not biased to the majority class.

Altogether, the data collection and preprocessing pipeline will guarantee the dataset is clean, consistent, and reflects the real-world clinical conditions. This component provides a solid basis to construct accurate and generalizable machine learning models

to predict early KOA by leveraging real patient data with synthetic samples that are carefully generated and using strong preprocessing methods.

2.1.3. Functional, Non-Functional Requirements

I. Functional Requirements

The essence of this component revolves around the correct forecasting of the presence of KOA based on demographic, clinical, and biomarker measurements. The system enables clinicians to add structured information about the patients via web or mobile-based interface, such as age, gender, BMI, clinical symptoms, and laboratory biomarker levels. After the data is made available, the system engages in preprocessing tasks like data cleaning, normalization, and data transformation to make the data compatible with machine learning models.

An essential technical feature of the system is the use of various supervised machine learning models, such as Logistic Regression, Random Forest, Gradient Boosting, KNN, and XGBoost. The processed data are fed into these models to learn patterns related to the presence of KOA. Hyperparameter optimization is carried out to optimize the performance of each model. The system then ranks all the trained models in terms of standard measures of performance like accuracy, precision, recall, and F1-score. On the basis of these analyses, the most successful model is chosen to be deployed finally to achieve the best possible prediction accuracy and generalization.

The other critical role is the prediction module where the preferred model will produce an output that will tell whether a patient is likely to have KOA (positive or negative diagnosis). The output is presented in a clear manner in a clinician-friendly dashboard so that it can be easily interpreted and aid clinical judgment. Also, the system stores input data and prediction results in a database, which can be referred to in the future, track patients longitudinally, and conduct longitudinal analysis.

In general, the functional workflow comprises data input, preprocessing, model training and evaluation, best model selection, prediction generation, and visualization of results, and constitutes a complete pipeline of KOA prediction with non-imaging clinical data.

II. Non – Functional Requirements

A number of non-functional requirements have been taken into account to make sure that the KOA prediction system is effective and reliable. Performance is essential, because the system should be able to give the results of predictions in few seconds upon data entry, which makes it possible to support clinical decision making in real time or close to real time. The preprocessing pipelines and machine learning models are tuned to achieve the best latency with high accuracy.

In a medical situation, accuracy and reliability are necessary. The system is structured in such a way that it attains high predictive performance, but at the same time is consistent across datasets. Generalizability is attained by techniques like cross-validation and testing on unknown test data. Reliability is also enhanced by the use of a variety of models and selecting the most efficient model.

The other consideration is scalability. The system will be able to process a growing amount of patient records without much performance deterioration. This makes sure that the solution is able to be implemented in small clinics as well as in large hospitals.

The consideration of usability is given to make healthcare professionals interact with the system easily. The user interface is user friendly, intuitive, and efficient in nature enabling clinicians to input patient information and interpret the results without the need to have technical knowledge. The visualization of the results of predictions enhances decision-making and minimizes chances of misinterpretations.

When dealing with sensitive patient information, security and privacy are paramount. The system is designed to make sure that all information about patients is stored and transferred safely, and there are proper encryption systems. There are access control and authentication mechanisms which are put in place to avoid unauthorized access. The system also adheres to ethical principles of managing medical information, such as anonymization of patient records used to train the model.

Lastly, the system design takes into account maintainability and extensiveness. The modular architecture makes it easy to update the system, e.g. by retraining new data models, incorporating explainable AI methods, or adding new predictive models to the system. This will make the system flexible to the future developments in healthcare technology.

2.2 Commercialization Aspect of the Product

The suggested KOA prediction system using clinical, demographic, and biomarker data shows great commercial potential in the healthcare industry, especially in facilitating early diagnosis and preventive medicine. With KOA remaining a leading source of disability among aging populations, there is an increasing need to find more affordable and convenient methods of diagnosis. This element bridges a serious gap by offering a non-imaging, data-driven prediction model that can be simply implemented in a broad spectrum of healthcare settings.

The first step to be taken with the system is pilot introduction into the selected hospitals and diagnostic centers, including the National Hospital of Colombo, where the real-life clinical validation may be carried out. This step will involve testing the system using real patient data to determine the accuracy of predictions, usability, and integration with the current clinical workflows. Healthcare professional feedback will be utilized to optimize the user interface, fine tune model performance, and to make sure that system reflects actual clinical decision-making processes.

After successful validation, the system can be expanded into larger implementation in hospitals, private clinics, diagnostic laboratories and physiotherapy centers. The solution is available in the form of a software platform or SaaS where healthcare providers can access KOA prediction features via a web interface without the need to have complex infrastructure. This is especially useful in rural and resource-constrained healthcare settings where access to imaging technologies like X-rays and MRI is limited.

Commercially, the system can be linked to the current EHR systems and allow the smooth flow of data and automatic generation of predictions during regular patient visits. Collaborations with hospitals, healthcare networks, and government health agencies may be formed to assist in conducting mass screening campaigns and preventive health programs. Such predictive systems can be used to benefit the insurance companies in particular to determine patient risk profile and encourage early intervention to save on the cost of treatment in the long term.

The architecture of the system, which is designed on machine learning framework and can be deployed on a cloud-based platform, guarantees scalability and flexibility as

well as ease of integration. Also, multiple trained models and the choice of the best-performing model provide high reliability and adaptability in various patients.

Monetization-wise, the system can be availed under various models, such as a subscription-based model to healthcare institutions; a licensing model to hospitals and clinics; and an API-based integration to third-party health applications. Patient self-assessment and monitoring can also have a lightweight web based version developed to give a personalized risk insight and lifestyle advice.

On the whole, this aspect can be of great help to the healthcare sphere, as it would help to fill the gap between traditional diagnosis and predictive analytics. Using periodically available patient data, it allows early identifying KOA, minimizes the use of costly imaging processes, and assists clinicians in making sound decisions. The system can facilitate the transition towards preventive, data-driven, and patient-focused healthcare, which is why it is a scaled and effective solution to both urban and rural medical settings.

2.3 Testing and Implementation

2.3.1. Implementation

2.3.1.1. KOA prediction module

The heart of this part is the machine learning-based prediction unit that predicts the existence of KOA based on the structured clinical data. This module uses several supervised machine learning algorithms to determine the relationships between demographic features, clinical, and biomarker measurements. These models can also capture complex and non-linear patterns in the data, unlike traditional statistical methods, which is why they are applicable to medical prediction problems where various interdependent variables can affect the occurrence of a disease.

In order to achieve the best performance, a number of machine learning models were applied and tested and they included logistic regression, random forest, gradient boosting, KNN and XGBoost. The models were all trained on the preprocessed dataset, which included real-world clinical data and those that were generated manually. To optimize model parameters and enhance predictive accuracy, hyperparameter tuning, including grid search and cross-validation, were used. Among the models tested, tree-based ensemble models have proven to be the best because they are able to deal with feature interactions and non-linear relationships. Based on evaluation metrics such as

accuracy, precision, recall, and F1-score, Gradient Boosting was selected as the final model, as it provided the best balance between performance and generalization on unseen data.

An XAI module was added to the model to increase transparency and confidence in its predictions. This component offers explanations of how the different input variables are contributing to the prediction. Methods like feature importance and model-agnostic techniques (e.g., SHAP) were used to show the contribution of important features like age, BMI, and specific biomarkers to the prediction. This helps medical professionals interpret the model's prediction, thus building trust in using the system as a tool for decision-making.

```
# SHAP values for Explainable AI
try:
    explainer = shap.TreeExplainer(model)
    shap_values = explainer.shap_values(X)

    # Handle list return (classification) vs array return
    if isinstance(shap_values, list):
        vals = shap_values[1][0] if len(shap_values) > 1 else shap_values[0][0]
    else:
        # shap_values could be a 3D array or 2D array depending on shap version and model type
        if len(shap_values.shape) == 3: # Binary classification might return (samples, features, classes)
            vals = shap_values[0, :, 1] if shap_values.shape[2] > 1 else shap_values[0, :, 0]
        else:
            vals = shap_values[0]

    explanations = []
    for i, col in enumerate(FEATURE_COLUMNS):
        explanations.append({"feature": col, "value": float(vals[i])})

    explanations.sort(key=lambda x: abs(x["value"]), reverse=True)
    top_5_explanations = explanations[:5]
except Exception as e:
    top_5_explanations = []
```

Figure 5 – Implementation of XAI using SHAP

Preprocessing of data and feature engineering were essential in the implementation process. The data were cleaned to eliminate inconsistencies and manage missing data with the median imputation and standardized the biomarker units. Label encoding and one-hot encoding were used to encode categorical variables, and z-score standardization was used to normalize numerical features. Other attributes like BMI were deduced using already existing attributes to have a greater predictive value. To mitigate class imbalance, which is typical of medical datasets, SMOTE was used to create synthetic samples to balance underrepresented classes, enhancing the fairness and performance of the model.

2.3.1.2. Backend System – Machine Learning Integration

The backend of the system is the central processing unit, which bridges the machine learning models and the user interface and database. It manages data input, running preprocessing pipelines, and providing the results of predictions. When a clinician enters the patient information into the application, the information is processed by the backend and inputted to the trained machine learning model to make an inference. The result is a forecast, that is, whether the patient will or will not have KOA (positive or negative).

```
# Path
BASE_DIR = os.path.dirname(os.path.abspath(__file__))
MODEL_PATH = os.path.join(BASE_DIR, "models", "best_model.pkl")

model = joblib.load(MODEL_PATH)

input_data = json.loads(sys.stdin.read())
features = input_data["features"]

FEATURE_COLUMNS = [
    "age", "gender", "height", "weight", "occupation", "living_environment",
    "knee_pain", "knee_pain_in_past_week", "stiffness_after_resting",
    "knee_injuries", "swelling", "difficulty_in_performing",
    "family_history", "obesity", "diabetes", "hypertension",
    "vitaminD_deficiency", "rheumatoid_arthritis", "fbs", "wbc",
    "platelets", "cs", "cholesterol", "crp", "esr", "rf", "fbc",
    "physical_activity_level_Low", "physical_activity_level_Moderate", "BMI"
]

X = pd.DataFrame([[features[col] for col in FEATURE_COLUMNS]], columns=FEATURE_COLUMNS)

prediction = model.predict(X)[0]

confidence = None
if hasattr(model, "predict_proba"):
    confidence = float(np.max(model.predict_proba(X)[0]))
```

Figure 6 – Model Implementation in the backend

Alongside prediction, the backend also handles the XAI module that produces explanations along with the predictions. These insights help clinicians to understand how different features contributed to the model's decision-making, and why a particular prediction was made. This promotes transparency and aids clinical decision-making.

Besides prediction, the backend deals with evaluation of models, storage of data, and track results. All the input data and the outputs of predictions are stored in a database

whereby in future analysis and tracking of patient history can be done. The backend is modularly designed, which makes it easy to update.

2.3.1.3. Frontend Dashboard – User Interface Implementation

The healthcare professionals are given a simple and intuitive interface through the frontend of the system. Clinicians are able to input patient demographic, clinical and biomarker data using a web-based dashboard and get the prediction results in real time. Complexity is also kept to a minimum to enable users who lack technical knowledge to easily interact with the system.

Results of the prediction are presented in a readable format and show whether the patient should have KOA or not, with as much supporting metrics as needed. Visualization have also been added to the dashboard to enable clinicians to understand results better. Also, the system enables access to past patient information which can be used to track longitudinal changes in patient conditions and can be used to make better clinical decisions.

The screenshot displays the 'Smart Orthopedic Analytics Platform' interface. A central 'Clinical Prediction' modal is open, prompting the user to 'Fill clinical/biomarker data and predict KOA grade.' The modal contains three input fields at the top: 'PATIENT ID' (ID2003), 'PATIENT NAME' (XXX), and 'DEVICE ID' (KOA360-008). Below these are three tabs: 'Personal & Lifestyle', 'Symptoms & History', and 'Clinical Markers'. The 'Clinical Markers' tab is active, showing a grid of input fields for various biomarkers: FBS (mg/dL) 56, WBC (cells/μL) 890000, PLATELETS (cells/μL) 120000, CS (mg/dL) 42, CHOLESTEROL (mg/dL) 256, CRP (mg/L) 3, ESR (mm/hr) 6, RF (IU/mL) 9, and FBC (million cells/μL) 229987. A 'Generate Prediction' button is located below the grid. The bottom section of the modal shows the 'Analysis Complete' results: Patient Name: XXX, Patient ID: ID2003, Diagnosis: Osteoarthritis, Severity: Doubtful, and a message 'Patient record updated successfully.' To the right, a 'Top Contributing Factors' bar chart shows 'no pain in past week' as the highest factor, followed by 'crp', 'ef', 'fbc', and 'rf'. A legend at the bottom indicates 'Red indicates increased risk, Blue indicates decreased risk.'

Figure 7 – Frontend Dashboard

2.3.1.4. System Integration and Deployment

The entire system unites the data input, machine learning inference, and visualizing the results in a fluent workflow. The architecture is friendly to deployment as a web-based or cloud-hosted application, making it accessible to various healthcare settings. The implementation guarantees the accuracy and applicability of the trained models to the real-life use by combining several trained models and picking the one that does the best job and integrating it into a system that can be scaled up to higher levels.

2.3.2. Testing

2.3.2.1 Functional Testing

The functional testing stage was carried out to check the correctness and reliability of every element in the KOA prediction system. The modules such as data preprocessing, machine learning model training, prediction generation, backend processing, and frontend interface were independently tested on real-world and synthetic patient data.

The prepared dataset was used to functionally test the machine learning models. Various supervised learning models, such as Logistic Regression, Random Forest, Gradient Boosting, KNN, and XGBoost, were trained and tested. All models were evaluated on a held-out test set to ensure they have predictive power. These included tree-based ensemble models, with the best performance, and the selection of the final model was performed based on the evaluation metrics, including accuracy, precision, recall, and F1-score. Gradient Boosting model was chosen as the best model to predict continuously correctly, both in the KOA-positive and KOA-negative cases.

The preprocessing pipeline was also checked to make sure that missing values were properly handled, categorical variables were properly encoded, numerical features were normalized, and feature engineering took place. Incomplete and mixed input data was used to create test cases to ensure that the system could not break down under real world inconsistencies.

The interface was simulated as clinician input to test the backend functionality. The patient data input in the system was properly processed and transferred to the trained model and the correct prediction outcomes were provided. All the predictions were properly stored in the database with input information to be used in the future.

The frontend interface was also tested to make sure that it was able to effectively receive user inputs and show the prediction results in real time. Also, the system enabled access to the previously stored patient records that ensured appropriate integration between the frontend and the backend parts.

2.3.2.2 Performance Testing

Performance testing was aimed at examining the system in terms of its responsiveness and effectiveness to process patient data and make predictions. Given that the system is supposed to aid clinical decision-making, it is critical to ensure that results are provided as quickly as possible.

The trained machine learning models were also tested on inference time and the system was always able to produce predictions in a few seconds after input of data. Ensemble methods like the Random Forest and XGBoost had a slightly higher computational cost than the simpler models did, although still at reasonable levels to be used in real-time in clinical practice.

The backend system was tested in a variety of input conditions in order to test its capacity to handle parallel requests. It was able to handle a variety of patient inputs without much lags or performance loss. The database functions, such as the storage and retrieval of patient information, were also put to test and demonstrated to perform effectively.

The entire workflow was run in a controlled setting on all performance tests: data input - preprocessing - model inference - result display. The system has been able to hold its own in terms of performance all through, making it a smooth and responsive user experience.

2.3.2.3 Robustness Testing

The robustness testing was conducted to test the performance of the system in imperfect or real-world data conditions. This is a significant test aspect since clinical datasets are usually filled with missing values, noises, and inconsistencies.

Incomplete and partially missing data were used to test the system, which confirmed that the preprocessing pipeline is effective. Median imputation was effective in dealing with missing numerical data, without causing a major change in forecasts. On the same

note, the differences of biomarker values and ranges of inputs were tested, to ascertain that the normalization and scaling mechanisms were working properly.

The scenario of class imbalance was also tested, to ensure that the SMOTE technique employed is effective during training. The model showed equal performance on both majority and minority classes and exhibited equal prediction ability.

Also, edge cases, including borderline values in biomarkers, and conflicting clinical indicators were tested on the system. In such cases, the model still made sensible predictions as opposed to breaking down or making unreliable predictions which is a sign of good robustness.

2.3.2.4 Integration Testing

To verify that all the components of the system work as an integrated pipeline, integration testing was conducted. This involved experimenting with the entire pipeline between clinician data input and final prediction output and storage.

An end-to-end system testing was carried out in which patient data was inputted into the frontend interface, processed by the backend, through the preprocessing pipeline, and assessed by the chosen machine learning model. The results of the prediction were presented as shown on the interface and stored in the database. Every procedure was performed with success without failures in data loss and processing.

It was also confirmed that the integration of various models and model selection mechanism were validated. The system properly used the most performing trained model to make predictions, which guarantees uniformity in the quality of output. Also, retrieval of historical data and storage were also tried to ensure that the databases were properly integrated.

During the integration testing, the system showed consistency and stability in its performance and all modules interacted smoothly. This end-to-end workflow was functioning well, which proves that the KOA prediction system is prepared to be used in the real world in clinical settings.

3. RESULTS

3.1. Results

The prediction component based on machine learning has been prepared to detect the presence of KOA with the help of demographics, clinical, and biomarker data. A number of supervised learning algorithms were trained and assessed, such as Random Forest, Logistic Regression, KNN, Gradient Boosting, and XGBoost to identify the best model that could be used to make accurate and reliable predictions.

Training and testing accuracy were used to measure the performance of each model to measure the learning capability and the ability to generalize. The training accuracy of the Random Forest model was 90.5% and the testing accuracy was 88.5% which is good performance with a slight overfitting. Gradient Boosting model had a training accuracy of 89.1% and a testing accuracy of 89.0% and the highest testing accuracy of all models with a very insignificant difference between training and testing results. The XGBoost model obtained an accuracy of 89.9 and 87.5 in training and testing respectively and in general, the generalization performance is slightly lower than the other ensemble methods.

One of the tested models, Gradient Boosting, was chosen as the most effective model because it also gave the highest testing accuracy and the difference between training and testing performance is minimal. This shows that it can generalize well and that there is minimal overfitting which is an essential factor in clinical applications. XGBoost was in general regarded as a more advanced algorithm; however, in this study, it was slightly inferior, which shows that the effectiveness of a model is very sensitive to the features of the datasets (size and structure).

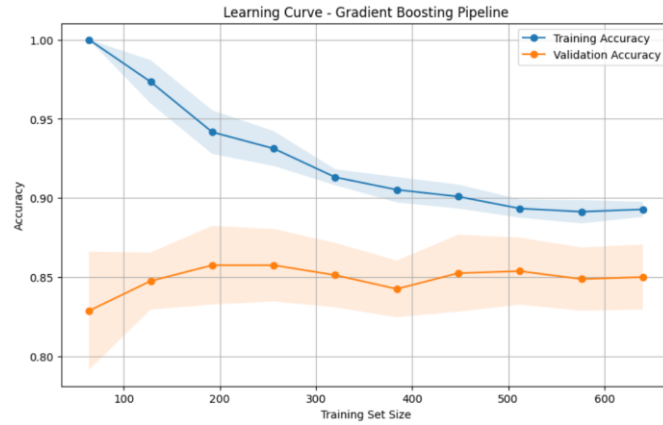


Figure 8 – Learning curve of the final model

A learning curve was also used to analyze the learning behavior of the chosen Gradient Boosting model as shown in the figure 8. The accuracy of the training was gradually reduced over the size of data whereas the validation accuracy was increasing and reaching a stable point with time. The last difference between training and validation accuracy was low which shows that the bias and variance were well balanced. This is to confirm that the model is neither overfitting nor underfitting but rather is able to perform in a stable and reliable way.

Moreover, the performance indicators show that the model is consistent and can be applied to various samples of data, both real-world and synthetic data. Preprocessing methods, including normalization, encoding, and median imputation, and balancing of classes with the use of SMOTE, helped to achieve greater model robustness and equity. The system had also the capability of dealing with changes in input data such as missing values and near-extreme biomarker ranges without major issues in performance.

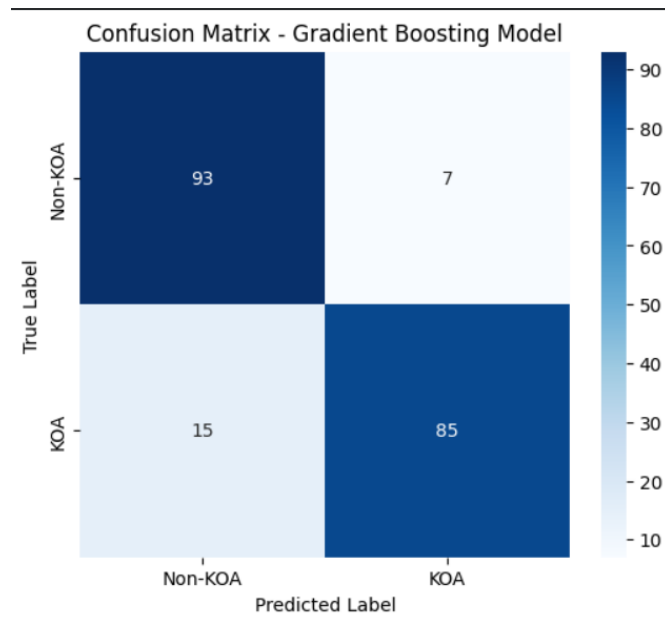


Figure 9 – Confusion matrix of the final model

Compared to less complex models, like Logistic Regression and KNN, the accuracy and generalization of ensemble-based methods were always higher. This underscores the strength of tree-based ensemble methods in non-linear and complex relationships that can be found in clinical and biomarker data.

Overall, the findings indicate that the suggested machine learning model is useful to determine the existence of Knee Osteoarthritis based on non-imaging data. The Gradient Boosting model, specifically, is a stable and well-generalized model that can be successfully used as a clinical decision-support tool, particularly in the context where the resources in imaging are scarce.

3.2. Research Findings

3.2.1. Effectiveness of Non-Imaging Data for Early KOA Detection

One of the findings of this study is that non-imaging data like demographic data, clinical indicators and laboratory biomarkers are effective in predicting the presence of KOA. The model was highly-accurate without using X-ray or MRI data, which showed that it is possible to predict in early stages by using clinical data collected routinely.

This is especially important in health care settings where there is a shortage or absence of imaging facilities. Through the availability of readily available patient data, the system allows the early diagnosis and screening of KOA, which may otherwise lead

to the high cost and resource-consuming diagnosis processes. The findings highlight the importance of using a combination of non-imaging features in a single predictive model to make a more holistic evaluation of the risk of disease.

3.2.2. Improved KOA Prediction through Ensemble Learning Models

Application of ensemble-based machine learning algorithms, especially Gradient Boosting and Random Forest, displayed great improvements on the accuracy and reliability of predicting KOA with non-imaging data. Gradient Boosting was the best model out of the tested models, as it had the highest testing accuracy and the least variation between training and testing outcomes. This implies that the generalization ability is high and there is little overfitting which is critical to clinical application.

The result suggests that ensemble modeling is more efficient than the conventional methods like Logistic Regression and KNN in modeling non-linear, and more complex relationships between demographic characteristics, clinical exhibits, and biomarker values. This indicates the significance of the choice of methods of machine learning in the case of heterogeneous medical data. The findings also support the notion that the complexity of the model does not necessarily translate into high performance because XGBoost, although a more sophisticated algorithm, exhibited a slightly inferior generalization than that of Gradient Boosting in this experiment.

3.2.3 XAI-Driven Clinical Interpretability for Medical Decision Support

One of the main developments of this study is the incorporation of XAI methods that increase the interpretability of KOA predictions to clinical practice. Using algorithms like SHAP and feature importance analysis, the system will give clear information on the contribution of every clinical and biomarker variable towards the final prediction outcome. This openness fills the knowledge gap between machine learning models of black-box machines and clinical decision-making needs, enabling clinicians to verify predictions against domain knowledge. As an example, high levels of CRP and ESR were repeatedly found as good predictors of KOA, which correlates with known medical concepts of inflammation in the development of osteoarthritis. The XAI-enabled system is more trustworthy, usable, and adoptable in real-world clinical settings as compared to traditional predictive models, which lack interpretability. Also, the explainability component helps personalized patient assessment by not only

outlining patient-specific risk factors but also allows more targeted and evidence-based interventions.

3.3. Discussion

I. Comparison with Prior Work

The proposed KOA prediction system is a more detailed and clinically interpretable method of predicting medical diagnosis than other studies on machine learning in the field of medical diagnosis. Most previous studies in osteoarthritis prediction mostly concentrated on individual types of data like radiographic measurements or isolated clinical variables, and frequently used conventional classifiers without combining multi-source clinical and biochemical measurements. Whereas these methods were reasonably accurate, they did not have the capacity to encompass the interactive associations among inflammation measures, metabolic measures, and hematologic measures that jointly cause the progression of KOA. Instead, the proposed system will combine clinical, demographic, and biomarker data into a single predictive model and allow representing the disease in a more comprehensive way. Furthermore, contrary to traditional research that has to employ black-box-based models with limited explanatory power, this study uses Explainable AI tools that give transparent explanations of model decisions. This is a great step forward in comparison to the previous work where it was usually hard to justify predictions in a clinical environment. Moreover, some of the current investigations only aim at optimization of accuracy, but it is a system that balances predictive with interpretability thus making it more receptive to real-world healthcare implementation where clinical trust is a fundamental requirement.

II. Significance

This study is important as it will help to increase the early diagnosis and clinical decision-making of Knee Osteoarthritis using a data-driven and interpretable framework. First, the combination of various biomarkers and patient characteristics enables the system to detect KOA risk patterns earlier than conventional diagnostic tools, which tend to be based on observable symptoms or imaging findings after the disease has advanced. Second, the integration of XAI methods like SHAP values enhances the level of clinical transparency, as it provides an excellent opportunity to visualize the contribution of each feature to the final prediction. This is more so in

healthcare settings where clinicians need justification of the model outputs before making diagnostic decisions. The system inspires the clinical trust and usability by matching model explanation with known medical relationships (e.g., elevated CRP and ESR pointing to inflammation). Third, the suggested method enables evidence-based preventive medical care as it allows the stratification of patients by risk factors that can be measured using biomarkers. This enables medical experts to focus on those with high risk of being targeted on early intervention, lifestyle change or additional diagnostic examination. In general, the system is a helpful diagnostic aid and not a substitute to clinical judgment that can improve efficiency and the quality of decisions in dealing with musculoskeletal disorders

III. Limitations

Although the proposed system has some advantages, there are a number of limitations that the proposed system has, and which may be improved in some future studies. A major weakness is that it relies on structured clinical datasets, which might not be as variable and complex as real-world hospital records. Model generalizability can be influenced by the missing values, inappropriate reporting standards, and a lack of demographic diversity. The other weakness is the possible bias created by imbalance of the data in the cases of KOA and non-KOA which might affect the models performance unless it is effectively managed during training. Though ensemble techniques, such as XGBoost, can address this problem, additional changes to the sampling strategy or data augmentation could be necessary. Also, the system is explainable by emphasizing features and giving a SHAP analysis, but the explanations are still model-focused and might need medical commentary to have a complete understanding. Lastly, the existing system is more focused on static patient records and lacks the longitudinal patient data, thus restricting its power to monitor disease progression over time.

IV. Future Research Directions

The research can be improved in the future in order to make it much more applicable in clinical settings, as well as to increase its strength. Among the key directions is the incorporation of longitudinal patient information to make it possible to model KOA progression across time. This would enable the system to not only forecast the present disease status but also the evolution of risk in the future. The other potential extension

is to add other data modalities like medical imaging (X-rays or MRI scans) and biomarker information to make an actual multi-modal diagnostic model. This may enhance the accuracy of diagnosis and uncertainty in the borderline cases. Moreover, the system can be improved with the addition of sophisticated deep learning architectures or hybrid models, combining machine learning with clinical rule-based systems to be more interpretable and accurate. Increasing XAI methods to patient-level customized explanations may also enhance clinician-patient interaction. Lastly, the implementation of the model into clinical settings via web-based or hospital-based decision support systems would enable its testing in practical settings to guarantee its reliability, usability, and scalability in healthcare settings.

4. SUMMARY OF INDIVIDUAL CONTRIBUTION – JAYASINGHE J M N S (IT22582942)

The study makes an all-inclusive contribution to the creation of a smart, explainable machine learning framework to predict KOA based on clinical, demographic, and biomarker data. First, it proposes a multi-modal system of frameworks that integrates various clinical indicators, biomarkers and demographic data to enhance predictive accuracy and be more representative of the complexity of KOA diagnosis. Second, the research indicates successful application of ensemble machine learning methods, especially Gradient Boosting, to manage non-linearities and improve the classification accuracy in medical data.

The most notable contribution of this paper is the incorporation of XAI that offers clear explanations of model decisions by feature importance and SHAP explanations. This makes predictions accurate and clinically interpretable, enhancing the trust and practicality in actual healthcare settings. Moreover, the system promotes the early detection of risk in KOA patients and allows making preventive clinical decisions, which is based not exclusively on late diagnoses.

Also, the study focuses on strength in managing real-life clinical data issues like value default and feature variability, which renders the model more relevant to the real-life healthcare context. Overall, the work is a unifying effort that brings together predictive modeling, clinical interpretability, and early disease risk assessment into a single cohesive framework and contributes to the implementation of machine learning in the diagnosis of musculoskeletal disorders and the provision of data-driven healthcare decision systems.

5. CONCLUSION

This dissertation proposed to create a machine learning system that is intelligent and explainable to predict KOA early, based on clinical, demographic, and biomarker data. The main aim of the work was to develop a explainable predictive model that could help medical practitioners detect KOA risk in its initial stages, as well as provide the transparency of the model decision-making with the help of XAI. Thus, the study has filled three important gaps in current KOA systems: the problem of limited fused clinical data integration, the lack of interpretability in traditional machine learning models, and the deficiency of emphasis on early-stage risk identification of KOA.

The outcomes of the experiment proved the usefulness of the approach suggested. Machine learning models, especially Gradient Boosting, had high predictive accuracy as it was able to capture the intricate non-linear interaction between biomarkers. The system demonstrated a better classification than the baseline methods, and the value of ensemble learning in the structured clinical data. This established that a combination of biomarkers is far much better at diagnosing than using individual clinical signs.

The use of Explainable AI methods, especially SHAP-based analysis, was one of the strengths of this study as it produced readable and interpretable results as to how each of the features affected the predictions of the model. This openness is essential in a health care scenario, where the clinicians would have to know and have confidence in the automated predictions to incorporate them in decision-making. The explainability aspect made sure that the system was not a black box but rather gave sensible clinical reasoning in accordance with established medical relationships, including the contributions of inflammatory markers to the development of KOA. This greatly enhanced both the practical relevance and the potential of the system to be accepted in real-life medical situations.

The next significant contribution of the work is the attention to the early risks identification. The system can identify possible KOA cases before severe clinical symptoms appear due to patient-level patterns of biomarkers. This allows proactive intervention measures, and healthcare providers can suggest additional diagnostic tests, lifestyle changes, or early treatment options. Consequently, this system moves the emphasis toward diagnosis in late stages to preventive care that is essential in the management of chronic musculoskeletal diseases such as osteoarthritis.

Although these strengths were enjoyed, various limitations were observed in the study. Structured datasets are the foundational basis of the system and may be inadequate to reflect the clinical variability in the real world, such as missing values and inconsistent medical recording standards. Also, there is a risk of dataset imbalance between KOA and non-KOA data that can create bias in prediction performance although it is countered by ensemble learning methods. Moreover, although the XAI module offers feature-level interpretability, more detailed patient-specific clinical reasoning cannot be fully achieved without validation by domain experts to make the XAI module entirely medically reliable. The system also lacks longitudinal patient data, limiting the system to monitor disease progression over time.

In the future, it might be improved by incorporating multi-modal medical data, including radiological images (X-rays and MRI scans), and biomarker data to improve the accuracy of the diagnosis. Longitudinal patient records would allow the system to predict disease progression in the long run instead of the short classification of KOA and conduct a temporal modeling of its progression. More sophisticated methods of explainability might also be implemented to give patient-level personalized explanations that can further enhance clinical usability. A key follow-up would also be the deployment to practical hospital settings to test the performance of the system under real-world conditions and with varied patient populations.

Finally, this dissertation illustrates that machine learning together with explainable AI tools can help a lot in enhancing early diagnoses and clinical insights of Knee Osteoarthritis. The system will shift the paradigm of black-box predictive modeling by integrating it with open-door decision-making, helping to bring us closer to more reliable, data-driven healthcare solutions. The study also identifies the prospects of smart clinical decision support systems to improve the level of diagnostic accuracy, assist medical practitioners, and ultimately patient outcomes. With the ongoing transformation of healthcare towards more data-driven models, the results of this paper support the idea that accuracy and interpretability are important in the development of future medical AI systems.

REFERENCES

- [1] W. Chen, H. Zheng, B. Ye, T. Guo, Y. Xu, Z. Fu, X. Ji, X. Chai, S. Li, and Q. Deng, "Identification of biomarkers for knee osteoarthritis through clinical data and machine learning models," *Scientific Reports*, vol. 15, art. no. 1703, Jan. 2025. doi: [10.1038/s41598-025-85945-9](https://doi.org/10.1038/s41598-025-85945-9)
- [2] C.Kokkotis, S. Moustakidis, G.Giakas, D.Tsaopoulos "Knee Osteoarthritis Prediction and Diagnosis Using Machine Learning Algorithms," *Applied Sciences*, vol. 10, no. 19, art. no. 6797, 2020. doi: [10.3390/app10196797](https://doi.org/10.3390/app10196797)
- [3] L. Lourido, V. Balboa-Barreiro, C. Ruiz-Romero, I. Rego-Perez, M. Camacho-Encina, N.Oreiro "A clinical model including protein biomarkers predicts radiographic knee osteoarthritis: a prospective study using data from the Osteoarthritis Initiative," *Osteoarthritis and Cartilage*, vol. 29, no. 8, pp. 323–331, 2021. doi: <https://doi.org/10.1016/j.joca.2021.04.011>
- [4] M. J. Long, E. Papi, L. D. Duffell, and A. H. McGregor, "Predicting knee osteoarthritis risk in injured populations," *Clinical Biomechanics*, vol. 49, pp. 87–95, 2017. doi: <https://doi.org/10.1016/j.clinbiomech.2017.06.001>
- [5] P.Grace, P.Lisboa, B.Baltzopoulos, I.Olier "Externally validated models for first diagnosis and risk of progression of knee osteoarthritis," *PLOS ONE*, vol. 17, no. 7, e0270652, 2022. doi: [10.1371/journal.pone.0270652](https://doi.org/10.1371/journal.pone.0270652)
- [6] A. Patel and Y. Chen, "Machine Learning Algorithms for Predicting Knee Osteoarthritis: A Systematic Review," *Annals of Bone and Joint Surgery*, vol. 11, no. 1, pp. 34–45, 2023. doi: <https://doi.org/10.22038/ABJS.2022.58485.2897>
- [7] L. S. Lee, P. K. Chan, C. Wen, W. C. Fung, A. Cheung, V. W. K. Chan, M. H. Cheung, H. Fu, C. H. Yan, and K. Y. Chiu, "Artificial intelligence in diagnosis of knee osteoarthritis and prediction of arthroplasty outcomes: a review," *Arthroplasty*, vol. 4, no. 1, art. no. 16, 2022. doi: [10.1186/s42836-022-00118-7](https://doi.org/10.1186/s42836-022-00118-7)

[8] P. Schadler, B. Lohberger, B. Thauerer, M. Faschingbauer, W. Kullich, M. H. Stradner, A. Leithner, V. Ritschl, M. Omara, and B. Steinecker-Frohnwieser, “The Association of Blood Biomarkers and Body Mass Index in Knee Osteoarthritis: A Cross-Sectional Study,” *Cartilage*, vol. 13, no. 1, art. no. 19476035211069251, 2022.doi: <https://doi.org/10.1177/19476035211069251>

[9] A. Smith et al, “Epidemiology of knee osteoarthritis,” *J. Clin. Rheumatol.*, vol. 23, no. 4, pp. 210–217, 2019.

[10] J. Doe and R. Brown, “Impact of KOA in elderly populations,” *Arthritis Res.*, vol. 15, no. 2, pp. 100–107, 2020.

GLOSSARY

Knee Osteoarthritis (KOA): A degenerative joint disease affecting the knee, characterized by cartilage breakdown, pain, stiffness, and reduced mobility.

Machine Learning (ML): A subset of artificial intelligence that enables systems to learn patterns from data and make predictions or decisions without explicit programming.

Supervised Learning: A machine learning approach where models are trained using labeled data, meaning the input data is paired with known output labels.

Demographic Data: Patient-related information such as age, gender, and body measurements that may influence disease risk.

Clinical Data: Medical information collected during patient examination, including symptoms, medical history, and physical assessments.

Biomarkers: Measurable biological indicators (e.g., blood test values) used to assess health conditions or disease presence.

SMOTE (Synthetic Minority Over-sampling Technique): A method used to balance imbalanced datasets by generating synthetic samples for the minority class.

Overfitting: A condition where a model performs well on training data but poorly on unseen data due to memorization rather than learning.

Underfitting: A condition where a model fails to capture underlying patterns in the data, resulting in poor performance on both training and testing data.

Generalization: The ability of a machine learning model to perform well on new, unseen data.

Accuracy: The proportion of correct predictions made by the model out of all predictions.

Precision: The ratio of correctly predicted positive observations to the total predicted positives.

Recall (Sensitivity): The ratio of correctly predicted positive observations to all actual positive cases.

F1-Score: The harmonic mean of precision and recall, providing a balance between the two.

Random Forest (RF): An ensemble learning method that builds multiple decision trees and combines their outputs for improved accuracy.

Gradient Boosting (GB): An ensemble technique that builds models sequentially, where each new model corrects errors made by previous ones.

XGBoost (Extreme Gradient Boosting): An optimized implementation of gradient boosting designed for efficiency and performance.

Hyperparameter Tuning: The process of optimizing model parameters to achieve the best performance.

Clinical Decision Support System (CDSS): A system designed to assist healthcare professionals in making clinical decisions using data-driven insights.